

Multi-Office Network Connectivity Report

Executive Summary

This project involved the design and deployment of a dual-office network architecture using Cisco Packet Tracer. The goal was to establish seamless communication between two geographically separated LANs (Office A and Office B) using RIP Dynamic Routing. By configuring Cisco routers via the Command Line Interface (CLI), I transitioned the hardware from a base state to a fully routed environment, ensuring all end-devices achieved full connectivity.

Technical Responses

Routers and Subnets

- **Why Routers?** Routers are required because Office A and Office B reside on different logical networks (subnets). While switches connect devices *within* a network, only a router can bridge the gap between *different* networks by making path decisions based on IP addresses.
- **The /30 WAN Link:** We used a **/30** mask for the router-to-router link because it is highly efficient, providing only 2 usable IP addresses. In a point-to-point connection, any more would be a waste of address space.

The Default Gateway

- **Purpose:** The Default Gateway acts as the "exit door" for a PC. When a PC wants to send data to an IP it doesn't recognize (like a PC in another office), it sends that data to the Gateway (the router) to handle the delivery.
- **Missing Gateway:** If the gateway is missing or incorrect, the PC can talk to its neighbors on the same switch but becomes a "ghost" to the outside world—it will drop any packets intended for other networks because it doesn't know where to send them.

Static vs. RIP Routing

- **Comparison:** Static routing is manually entered and highly secure, but it doesn't scale well. RIP (Routing Information Protocol) is a dynamic protocol that allows routers to automatically share their routing tables with neighbors.

- Why RIP? RIP was the better choice for this lab because it automates the discovery of new networks. Once I configured **version 2** and the **network** commands, the routers "talked" to each other to build the bridge without me having to manually map every path.

Verification Tools

- **show ip interface brief:** This command is the "heartbeat" check for a network engineer. It confirms the IP address assigned to each port and, more importantly, shows the Status (physical link) and Protocol (logical link) are both "up/up".
- **DNS:** A DNS server is not required here because we are testing connectivity using raw IP addresses. DNS is only needed when we want to translate human-friendly names (like <https://www.google.com/search?q=google.com>) into IPs.
- **Simulation Tool:** This allows us to visualize the "invisible" flow of data. It helps us see the difference between ICMP (the actual ping) and ARP (the router finding the hardware address), allowing us to identify exactly where a packet gets dropped (the "blinking X").

IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Gateway
Router0	G0/0/0 (LAN)	192.168.1.1	255.255.255.0	N/A
Router0	G0/0/1 (WAN)	10.0.0.1	255.255.255.252	N/A
Router1	G0/0/0 (LAN)	192.168.2.1	255.255.255.0	N/A
Router1	G0/0/1 (WAN)	10.0.0.2	255.255.255.252	N/A

PC0	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.2.10	255.255.255.0	192.168.2.1

```
C:\>ping 192.168.2.10
```

```
Pinging 192.168.2.10 with 32 bytes of data:
```

```
Reply from 192.168.2.10: bytes=32 time=2ms TTL=126
```

```
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
```

```
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
```

```
Reply from 192.168.2.10: bytes=32 time<1ms TTL=126
```

```
Ping statistics for 192.168.2.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

```
Router(config-router)#do show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
```

```
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
```

```
       * - candidate default, U - per-user static route, o - ODR
```

```
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

```
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0/1
```

```
L       10.0.0.1/32 is directly connected, GigabitEthernet0/0/1
```

```
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
```

```
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
```

```
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
```

```
R       192.168.2.0/24 [120/1] via 10.0.0.2, 00:00:07, GigabitEthernet0/0/1
```

```
C:\>ping 192.168.2.11

Pinging 192.168.2.11 with 32 bytes of data:

Reply from 192.168.2.11: bytes=32 time<1ms TTL=126
Reply from 192.168.2.11: bytes=32 time<1ms TTL=126
Reply from 192.168.2.11: bytes=32 time<1ms TTL=126
Reply from 192.168.2.11: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

CoryMorris.pkt file

<https://drive.google.com/file/d/1r1JliUdiloFUSjcKZf35RZSt6s6XZDyG/view?usp=s>
haring